Design of an External Pull-Up Resistor for Port 0 of the 8051 Microcontroller

Tou Ka Wai
DC42557
University of Macau
ECEN1004

***Abstract*—This report analyzes the internal structure of Port 0 (P0) of the 8051 microcontroller and derives a single external pull-up resistor value for P0 used as a general-purpose I/O port. The derivation is based directly on the DC characteristics in the Intel 8051 data sheet and on clearly stated assumptions appropriate for a low-speed laboratory assignment.**

I. Introduction

In the standard 8051 architecture, Port 0 (P0) is a bidirectional port that is also multiplexed with the low-order address/data bus (AD0–AD7) for external memory access [1], [2]. Unlike Ports 1–3, P0 implements an open-drain output stage and does not provide a reliable internal pull-up when used as general-purpose I/O [1], [3]. Therefore, an external pull-up resistor is required between each P0 pin and V_{CC}.

The goal of this assignment is to: (i) review the internal structure of a single P0 bit, (ii) explain why an external pull-up is still necessary even though a transistor towards V_{CC} appears in the internal diagram, and (iii) use the Intel 8051 data sheet to derive a realistic pull-up resistor value for low-speed GPIO use of P0.

II. Internal Structure of Port 0

Fig. 2 shows a simplified internal structure of one Port 0 bit, P0.x, based on the manufacturer's documentation [1], [3]. The bit latch (P0.X LATCH) can be written from the internal data bus and read back through the *Read Latch* path. A separate *Read Pin* path allows the CPU to sample the actual voltage at the pin.

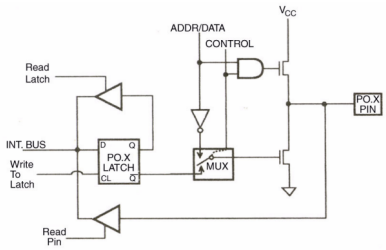

Fig. 1. Simplified internal circuit of one Port 0 bit (P0.x) [1].

The output stage includes a multiplexer (MUX) and an n-channel MOS transistor that connects the pin to ground when turned on [1], [3]. When this transistor is *on*, P0.x is actively driven low. When it is *off*, the pin is left in a high-impedance state. In normal GPIO mode there is no guaranteed, user-controlled path from the pin to V_{CC}, so a valid logic-high level cannot be achieved without an external pull-up resistor [1], [2].

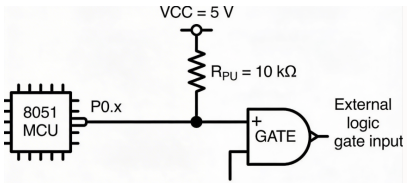

Fig. 2. External pull-up resistor for P0.x used as GPIO.

III. External Pull-Up Connection

To use P0.x as a regular input or output pin, an external pull-up resistor R_{PU} is connected between the pin and the supply voltage V_{CC}, as illustrated in Fig. 2 [3].

When the internal nMOS is off, the pin is left in a high-impedance state and R_{PU} pulls the node up to a high level close to V_{CC}. When the nMOS is on, the pin is driven low and current flows from V_{CC} through R_{PU} into the pin and then to ground.

IV. Why an Internal Transistor Cannot Replace the External Pull-Up

During discussions with classmates, we initially misinterpreted the internal diagram of P0. Since a transistor connected towards V_{CC} appears in the block diagram, it was tempting to treat it as a programmable pull-up element that could be controlled by software. If this were true, an external resistor might not be needed.

However, a closer inspection of the Intel data sheet shows that this interpretation is incorrect:

- The port description explicitly states that P0 is an *open-drain* port and that there is *no internal pull-up* for normal I/O operation [1], [3]. A valid logic-high level must be provided by an external resistor.
- The gate of the upper transistor is driven by the internal bus control logic during external memory cycles, not by a simple port configuration bit [1]. Its behaviour follows the AD0–AD7 timing and is not documented as a user-adjustable pull-up.

- In the DC characteristics, there is no IOH or pull-up specification for P0 that would correspond to a well-defined internal resistor; all VIH and VOH guarantees assume that appropriate external pull-ups are used [1], [2].

Therefore, for design and for this assignment, Port 0 must be modelled as an open-drain output that can only pull the pin low or release it. The external resistor between P0.x and V_{CC} is the only reliable pull-up element.

V. Deriving the Allowed Range of R_{PU} from the Data Sheet

In this section all numerical parameters are taken directly from the Intel 8051 data sheet for Port 0 [1], and every additional assumption is stated explicitly.

A. Lower Bound from the Sink Current Limit

The DC characteristics specify, for Port 0, an output low voltage

$$V_{OL1,\max} = 0.60 \text{ V} \quad \text{at} \quad I_{OL} = 3.2 \text{ mA},$$

for the worst case considered in the table [1]. We interpret $I_{OL} = 3.2$ mA as the maximum recommended low-level output current for a single P0 pin.

When P0.x outputs a logic low, the pull-up resistor is effectively connected between V_{CC} and ground, so the sink current is

$$I_{\text{sink}} = \frac{V_{CC} - V_{OL1}}{R_{\text{PU}}}. \tag{1}$$

To avoid exceeding I_{OL}, we require $I_{\text{sink}} \leq 3.2$ mA, which yields the formal lower bound

$$R_{\text{PU}} \geq R_{\text{PU,min}} = \frac{V_{CC} - V_{OL1}}{I_{OL}}. \tag{2}$$

Assuming $V_{CC} = 5$ V, this gives

$$R_{\text{PU,min}} \approx \frac{5 - 0.60}{3.2 \text{ mA}} \approx 1.4 \text{ k}\Omega. \tag{3}$$

B. Upper Bound from Input Leakage and VIH

For Port 0, the same table lists an input leakage current $|I_{LI}| \leq 100$ μA and a high-level threshold for port inputs of approximately $V_{IH} \approx 2.4$ V [1]. We assume that a P0 pin used as a digital input should satisfy

$$V_{\text{node}} \geq V_{IH} \tag{4}$$

even in the worst case of $I_{LI} = 100$ μA.

Under this assumption, the node voltage is

$$V_{\text{node}} = V_{CC} - I_{LI} R_{\text{PU}} \geq V_{IH}, \tag{5}$$

which leads to the upper bound

$$R_{\text{PU}} \leq R_{\text{PU,max}} = \frac{V_{CC} - V_{IH}}{I_{LI}}. \tag{6}$$

With $V_{CC} = 5$ V, $V_{IH} = 2.4$ V, and $I_{LI} = 100$ μA, we obtain

$$R_{\text{PU,max}} \approx \frac{5 - 2.4}{100 \ \mu\text{A}} = 26 \text{ k}\Omega. \tag{7}$$

Combining both inequalities, the data sheet implies that any resistor in the interval

$$1.4 \text{ k}\Omega \lesssim R_{\text{PU}} \lesssim 26 \text{ k}\Omega \tag{8}$$

will satisfy the worst-case DC limits for Port 0 at $V_{CC} = 5$ V.

VI. Choosing a Single Practical Value

The interval 1.4–26 kΩ is relatively wide. For a low-speed laboratory assignment where P0 is used mainly as GPIO, it is reasonable to choose a single standard resistor value inside this interval.

In this report we make one additional, explicit design preference:

- We prefer to keep the sink current well below the 3.2 mA limit, in order to reduce power consumption and provide extra safety margin in a student experiment.

Using the approximation

$$I_{\text{sink}} \approx \frac{V_{CC} - V_{OL1}}{R_{\text{PU}}}, \tag{9}$$

and still taking $V_{CC} = 5$ V, $V_{OL1} = 0.60$ V, we can evaluate the current for a candidate value $R_{\text{PU}} = 10$ kΩ:

$$I_{\text{sink}} \approx \frac{5 - 0.60}{10 \text{ k}\Omega} \approx 0.44 \text{ mA}. \tag{10}$$

This current is almost one order of magnitude smaller than the 3.2 mA rating, yet still large enough to make the node insensitive to microampere-level leakage currents [1]. At the same time, 10 k$\Omega \ll$ 26 kΩ, so the high-level constraint is comfortably satisfied.

Based on these considerations, this report fixes a single pull-up resistor value for all P0 pins:

$$R_{\text{PU}} = 10 \text{ k}\Omega. \tag{11}$$

This value lies strictly inside the theoretical interval [1.4 kΩ, 26 kΩ] derived from the data sheet and can be implemented with a commonly available E24 series resistor.

VII. Brief Note on Bus Operation

When P0 is reused as the multiplexed address/data bus (AD0–AD7), the bus capacitance and operating frequency are higher than in simple GPIO use. A more detailed analysis would then include the RC rise time $t_r \approx 2.2 R_{\text{PU}} C_{\text{BUS}}$ and the required bus timing [2]. In this assignment we restrict attention to low-speed experiments, where $R_{\text{PU}} = 10$ kΩ is still adequate for P0 even when it briefly acts as a bus line.

VIII. Conclusion

Using only parameters taken directly from the Intel 8051 data sheet and clearly stated assumptions, this report has derived an allowed range 1.4 k$\Omega \leq R_{\text{PU}} \leq$ 26 kΩ for the external pull-up resistor of Port 0 at $V_{CC} = 5$ V. Within this range, the value $R_{\text{PU}} = 10$ kΩ was selected as a practical choice for low-speed GPIO experiments, because it significantly reduces the sink current compared to the 3.2 mA rating while still guaranteeing valid logic levels.

REFERENCES

[1] *Intel 8051 Microcontroller Data Sheet*, Intel Corporation, 1988, available at https://www.alldatasheet.com/datasheet-pdf/view/107780/INTEL/8051.html.

[2] K. J. Ayala, *The 8051 Microcontroller*, 3rd ed. Cengage Learning, 2004.

[3] M. T. Inc., "Port 0 structure and open-drain operation in 8051-compatible mcus," https://www.microchip.com/, 2020.